

8086 CPU Architecture & Internal Registers

Q. Explain the internal architecture of the 8086 microprocessor. Describe the functions of the Bus Interface Unit (BIU) and the Execution Unit (EU).

> **Definition to Remember**

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(deco

1. Bus Interface Unit (BIU)

The BIU handles all data and address transfers on the system buses.

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2. Execution Unit (EU)

The EU is responsible for decoding and executing the fetched instructions.

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3. Parallel Operation

Because the BIU and EU operate independently, the BIU continuously fills the 6-byte queue from memory while the EU executes the previously fetched instructions. This prevents the EU from sitting idle waiting for memory fetches, drastically reducing execution time.

> **Must-Write Points (for 10 marks)**

> 1. 80

> **Quick Recall**

> `808
Paralle